

MSE211: Computational Method

Spring 2024 Project

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Introduction

The primary objective of this investigative project is to comprehensively model and simulate the oscillatory behavior of a pendulum, incorporating the effects of friction using both analytical and

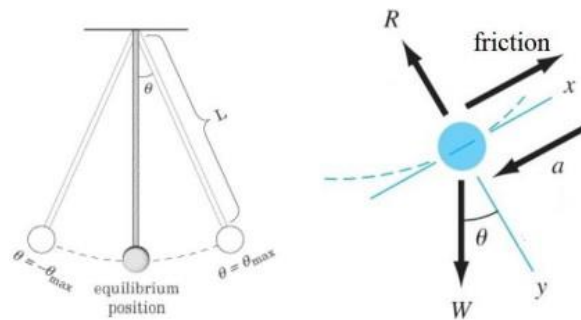


Figure 1: Free body diagram of a simple pendulum

Distinctively extending the explorations of part 2 of lab 4, this project differentiates itself by the inclusion of frictional forces in the analysis. By methodically applying the forces in equilibrium conditions along the x and y axes and embedding the frictional force into the equation through Newton's second law, we derive a second-order differential equation (ODE). This ODE is further simplified under the assumption that the angle θ remains small, which allows for a more tractable analytical expression of the pendulum's position θ as a function of time.

The scope of this project encompasses a detailed evaluation, analysis, and comparison of the numerical results obtained through the fourth-order Runge-Kutta method and the theoretical analytical solutions provided by the simplified models.

Methodology

We initiated our study by selecting three potential starting angles (θ) for the pendulum, all of which were constrained to be under 20° . The angles chosen were as follows:

θ_0	Initial Angle
θ_1	4°
θ_2	12°
θ_3	18°

Table 1: Selected Initial Angles

Our analysis commenced with an assessment of the cumulative forces acting along the x and y axes. The system is influenced by the tension in the string, gravitational pull, and frictional resistance. The gravitational influence is decomposed into a vertical component, which is exactly opposed by the tension in the string, and a horizontal component that contributes to the pendulum's motion:

$$F_g = -mg \sin(\theta)$$

$$F_f = -\lambda \frac{ds}{dt}$$

Here, F_g represents the horizontal gravitational force, and F_f the frictional force with λ being the coefficient of friction, s the tangential displacement, m the mass, and g the acceleration due to gravity.

By implementing Newton's second law, the dynamics are encapsulated by:

$$m \frac{d^2s}{dt^2} = -mg \sin(\theta) - \lambda \frac{ds}{dt}$$

Given that the displacement s can be expressed as $s = r\theta$ and thus $\frac{ds}{dt} = r \frac{d\theta}{dt}$, the motion equation simplifies to:

$$m \frac{d^2s}{dt^2} = -mg \sin(\theta) - \lambda r \frac{d\theta}{dt}$$

If we assume θ is sufficiently small to approximate $\sin(\theta) \approx \theta$, we redefine the equation as:

$$r \frac{d^2\theta}{dt^2} + \lambda r \frac{d\theta}{dt} + mg\theta = 0$$

This setup leads to two main differential equations used throughout this lab, labeled as equations 4 and 6. Specifically, equation 6, valid only for small θ , is modeled to predict oscillatory behavior:

$$\theta(t) = \exp(-\lambda t)(A \cos(\alpha t) + B \sin(\alpha t))$$

where A and B are constants to be determined under initial conditions where the pendulum starts from rest. This is expressed as equation 11:

$$\theta(t) = \theta_0 \exp(-\lambda t) \left(\cos(\alpha t) + \frac{\lambda}{2m\alpha} \sin(\alpha t) \right)$$

The constants and their relationships are further explored to define the pendulum's motion over time accurately.

Table 2: Assumptions on Pendulum Properties

Property	Value
m (Mass of the pendulum)	0.3 kg
r (Distance to center of gravity)	1 m
λ (Friction coefficient)	0.02 to 0.1

Using the fourth-order Runge-Kutta method, we analyzed how θ evolves under the influence of different friction coefficients (0.06, 0.07, 0.08, 0.09) and initial conditions using both equations 4 and 6 for a comprehensive simulation over 8 seconds.

The coding part of our analysis was conducted in MATLAB, where we developed scripts to simulate and plot the pendulum's displacement and velocity under varying conditions. These scripts employ standard MATLAB functions for computation and graphing, including linspace, sqrt, plot, and legend, with detailed annotations provided in the appendix.

In-depth computational routines were constructed to iteratively solve these equations using the designated numerical method, allowing us to visually compare the impact of different starting angles and damping coefficients on the pendulum's behavior. The generated plots distinctly illustrate the amplitude's decay over time, a manifestation of the damping effects caused by the friction in the system.

Results and Discussion

1. Displacement analysis

Our preliminary objective was to monitor changes in the pendulum's displacement over an 8-second duration, initiating from a static state. This analysis was conducted using MATLAB to animate the pendulum's motion via the specified differential equations. Typically, the graphical representation of angular displacement begins at the pendulum's highest initial amplitude (θ_0), which corresponds to its release position. As the experiment progresses, this amplitude diminishes due to energy losses primarily attributed to frictional forces within the system. These forces act to decelerate the pendulum, gradually reducing its motion until it ultimately ceases.

For a detailed exploration of the pendulum's motion, we engaged MATLAB to visualize the oscillation curve using both numerical solutions (Equations 4 and 6) and the analytic approach (Equation 11). This comprehensive method allows us to observe and analyze the pendulum's behavior under simulated friction conditions accurately.

The position of the pendulum as predicted by Equation 11 is described by the formula:

$$\theta(t) = \theta_0 \exp(-\lambda t) \left(\cos(\alpha t) + \frac{\lambda}{2m\alpha} \sin(\alpha t) \right)$$

In accordance with the lab manual, we were instructed to select experimental parameters that adhere strictly to the project's assumptions: a mass (m) of 0.3 kg, a radius (r) of 1 m, and a friction coefficient (λ) ranging from 0.06 to 0.09. Additionally, the initial angles (θ_0) were chosen to be less than 20 degrees—specifically 4°, 12°, and 18°.

Under these conditions, we utilized MATLAB to generate a series of plots illustrating the pendulum's positional changes throughout the 8-second interval. These visualizations provide a clear depiction of how the pendulum's path is influenced by the varying levels of friction imposed by the experimental setup.

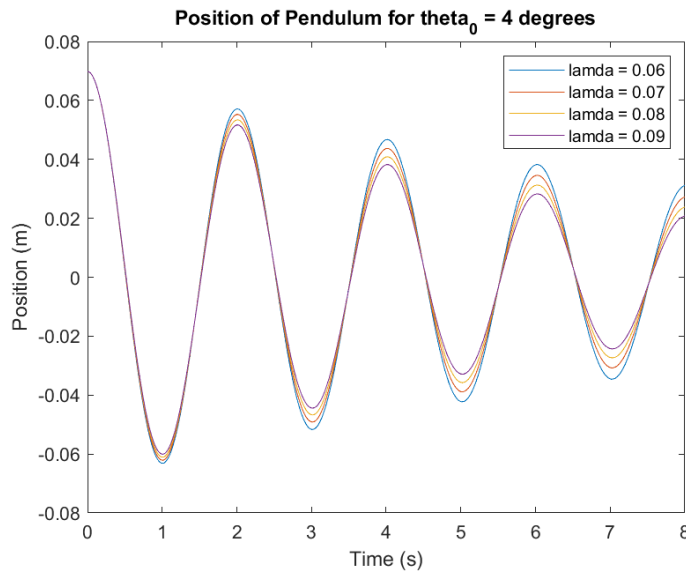


Figure 2: Position of the pendulum with $\theta = 4$

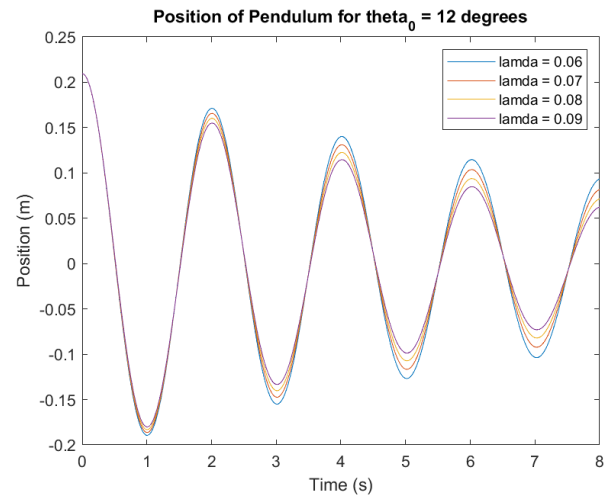


Figure 3: Position of the pendulum with $\theta = 12$

Observations from the MATLAB simulations clearly indicated that the pendulum's amplitude of oscillation is inversely proportional to the friction coefficient λ . With a lower λ , the pendulum experiences less resistance, allowing it to swing further from its initial position. This suggests that smoother paths (lower friction) enable the pendulum to achieve greater displacements.

In comparing the three generated graphs corresponding to different initial angles θ_0 of 4°, 12°, and 18°, it was evident that while the general shape of the displacement curve remained consistent, the amplitude increased with larger initial angles. This demonstrates that the pendulum reaches further extents when started from a higher displacement,

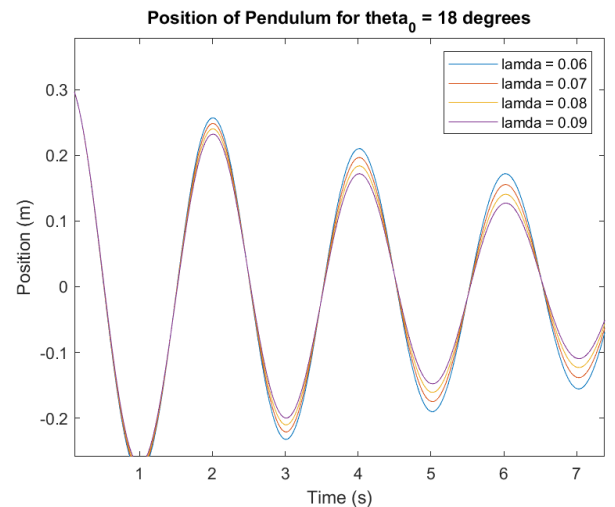


Figure 4: Position of the pendulum with $\theta = 18$

irrespective of the friction conditions.

To further analyze these behaviors quantitatively, we utilized a MATLAB function based on the following formula from our lab manual, which applies the Runge-Kutta 4th-order method for second-order differential equations:

Definition of Runge-Kutta 4th -order method for 2nd order ODE:

$$\begin{aligned}
 \frac{d^2x}{dt^2} &= f(t, x, v), \quad \frac{dx}{dt} = v, \quad x(t_0) = x_0, \quad v(t_0) = v_0 \\
 t_{i+1} &= t_i + h, \quad h = \text{stepsize} \\
 dx_1 &= h \times v_i \\
 dv_1 &= h \times f(t_i, x_i, v_i) \\
 dx_2 &= h \left(v_i + \frac{1}{2} dv_1 \right) \\
 dv_2 &= h \times f \left(t_i + \frac{1}{2} h, x_i + \frac{1}{2} dx_1, v_i + \frac{1}{2} dv_1 \right) \\
 dx_3 &= h \left(v_i + \frac{1}{2} dv_2 \right) \\
 dv_3 &= h \times f \left(t_i + \frac{1}{2} h, x_i + \frac{1}{2} dx_2, v_i + \frac{1}{2} dv_2 \right) \\
 dx_4 &= h(v_i + dv_3) \\
 dv_4 &= h \times f(t_i + h, x_i + dx_3, v_i + dv_3) \\
 x_{i+1} &= x_i + \frac{1}{6}(dx_1 + 2dx_2 + 2dx_3 + dx_4) \\
 v_{i+1} &= v_i + \frac{1}{6}(dv_1 + 2dv_2 + 2dv_3 + dv_4)
 \end{aligned}$$

Figure 5: Definition of the Runge-Kutta 4th-order method for 2nd-order ODE (Lab Manual)

For a comprehensive study, we proceeded to animate the pendulum's motion over 8 seconds using four different friction coefficients λ : 0.06, 0.07, 0.08, and 0.09. This range was chosen to align with the project's predefined conditions. By employing the Runge-Kutta 4th-order method outlined above, we plotted the oscillatory curves using both Equation 4 and Equation 6. These equations model the pendulum's motion under varying degrees of damping caused by friction:

$$\frac{d^2\theta}{dt^2} + \frac{\lambda}{m} \frac{d\theta}{dt} + g \sin(\theta) = 0$$

The resulting graphs illustrate how the pendulum's angular displacement evolves over time, providing a visual representation of the damping effects as influenced by different levels of friction. Through these plots, we could visually confirm the theoretical predictions regarding amplitude behaviors under different frictional conditions.

Additional Analysis Using Equation 6

Utilizing Equation 6, which is succinctly expressed as:

$$m \frac{d^2\theta}{dt^2} + k\theta = 0$$

where $k = mg$ (representing the combined effect of mass and gravitational force), we carried out further simulations to observe the pendulum's behavior under different initial conditions. These simulations were visualized through a series of MATLAB plots, each illustrating how the pendulum's angular displacement varies over time.

The following graphs were generated from this analysis:

- **Figure 9:** Angular displacement over time for $\theta_0 = 4^\circ$
- **Figure 10:** Angular displacement over time for $\theta_0 = 12^\circ$
- **Figure 11:** Angular displacement over time for $\theta_0 = 18^\circ$

Each graph showcases the trajectory of the pendulum's swing, allowing us to observe the natural oscillation pattern as predicted by Equation 6 under minimal frictional influence.

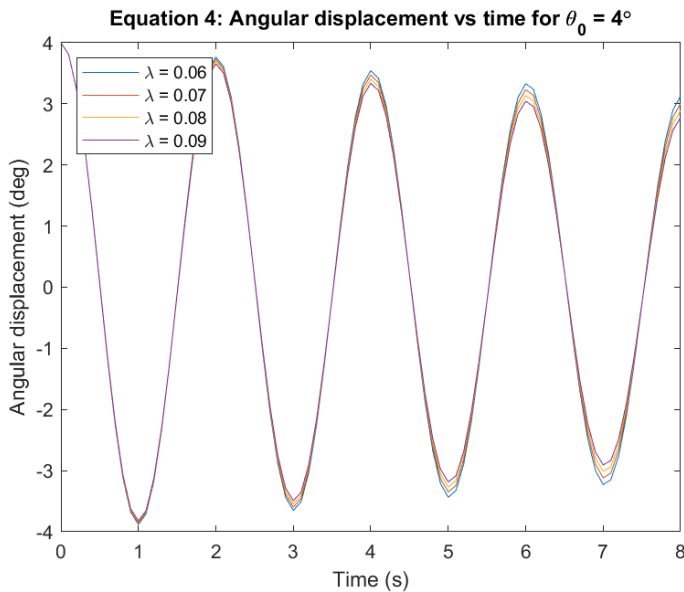


Figure 6: Angular displacement vs time using Equation 4 ($\theta = 4$)

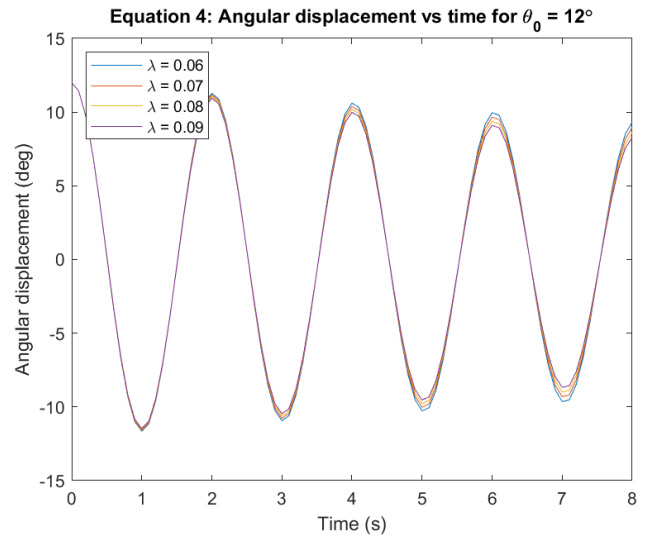
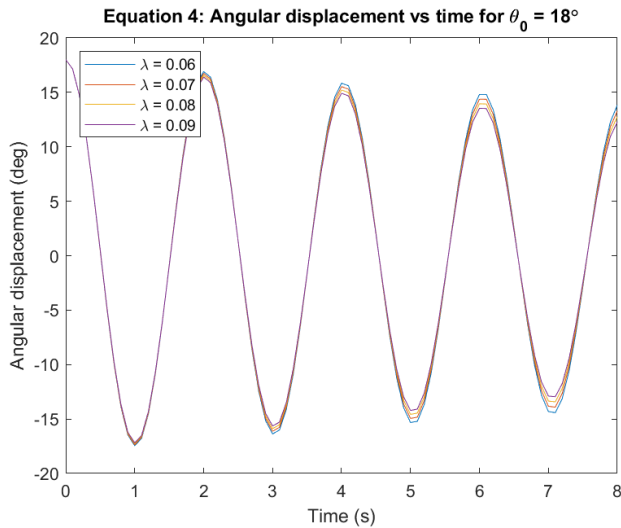


Figure 7: Angular displacement vs time using Equation 4 ($\theta = 12$)



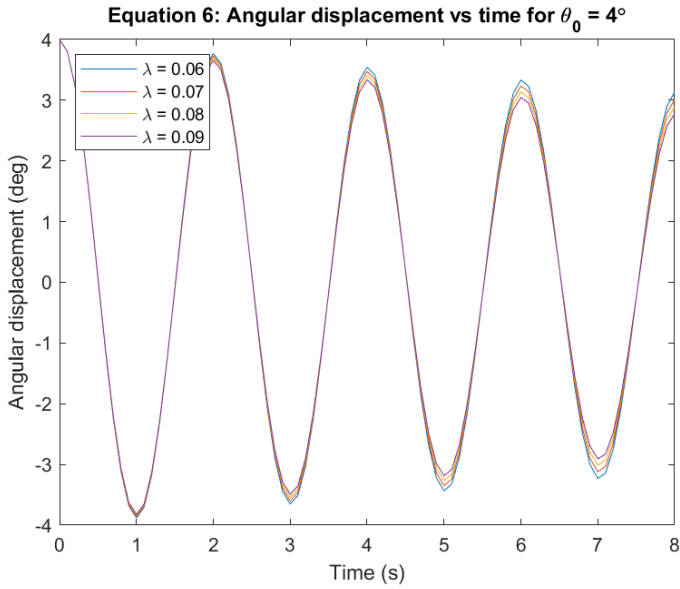


Figure 9: Angular displacement vs time using Equation 6 ($\theta = 4$)

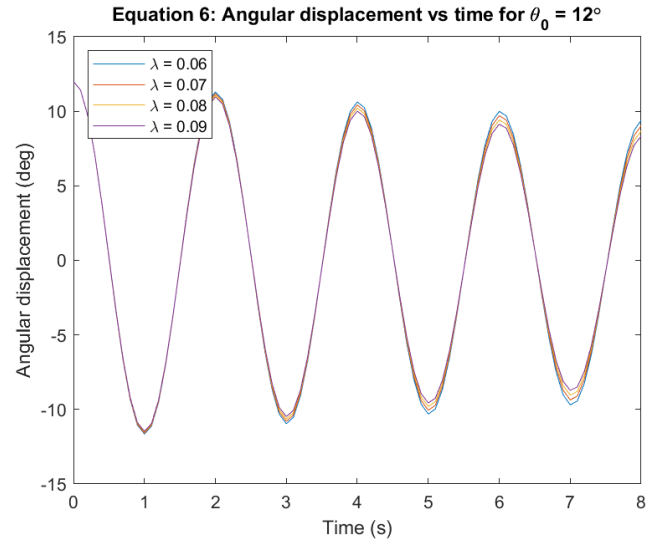


Figure 10: Angular displacement vs time using Equation 6 ($\theta = 12$)

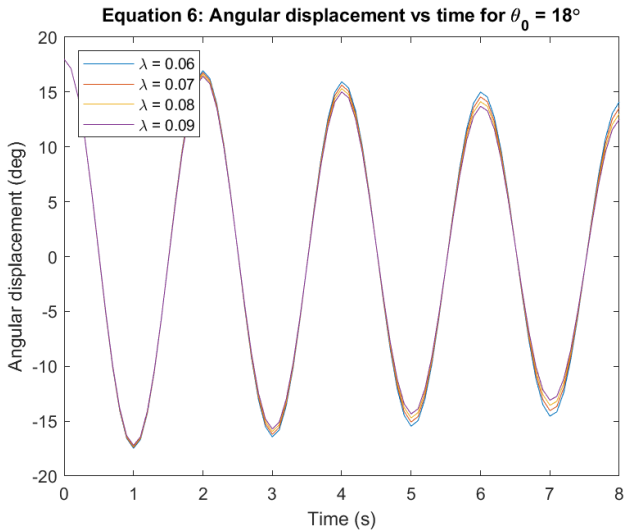


Figure 11: Angular displacement vs time using Equation 6 ($\theta = 18$)

Comparative Analysis of Displacement Graphs

Our analysis of the graphs generated from Equation 4 and Equation 6 indicates minimal variance in the shape of the curves, suggesting that changes in amplitude are the primary differences observed under these conditions. When examining the angular displacement over time for different initial angles (θ_0)—specifically 4° , 12° , and 18° —it becomes evident that the overall contour of the displacement graph remains consistent across different setups.

Key Observations:

- **Stability in Graph Shape:** Regardless of the initial displacement θ_0 , the fundamental shape of the angular displacement-time graph does not vary significantly. This uniformity underscores the robustness of the pendulum's oscillatory nature against changes in starting amplitude.
- **Influence of Friction Coefficient:** A lower friction coefficient (λ) generally results in a larger amplitude of motion. This observation aligns with physical intuition, as reduced frictional damping allows the pendulum to swing more freely and reach a wider arc.
- **Amplitude Variability:** While the shape of the graph remains stable, the amplitude clearly changes with different initial displacements. Each increment in θ_0 leads to a noticeable increase in amplitude, confirming that the pendulum's range of motion is directly proportional to its initial potential energy.
- **Delayed Effects of Friction:** Although initially subtle, the impact of varying friction coefficients on the amplitude becomes more pronounced over time. This gradual divergence highlights how friction progressively influences the energy dissipation within the system.

These findings suggest that while the basic oscillatory pattern of the pendulum is preserved under different theoretical conditions modeled by Equations 4 and 6, the extent of its swing and its damping behavior can vary significantly depending on both the initial angle and the friction coefficient. The consistency in graph shape across different tests provides a reliable baseline for further experimental validation and theoretical analysis.

Figure 13: Angular velocity vs time using Equation 11 ($\theta = 12$)

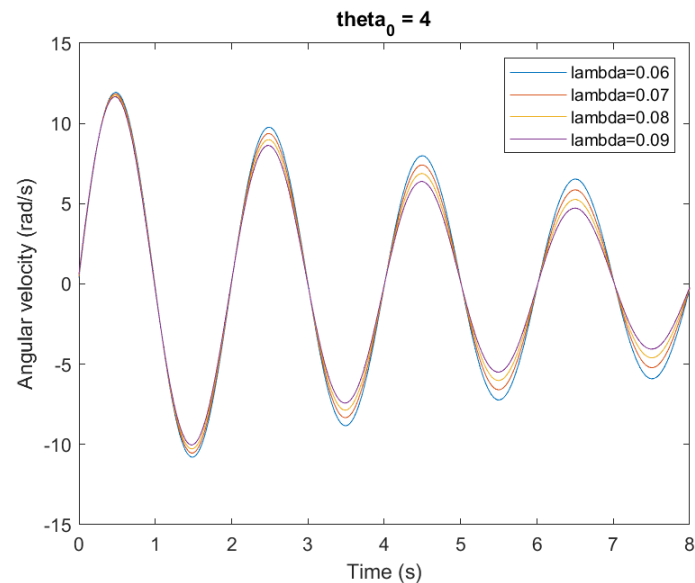
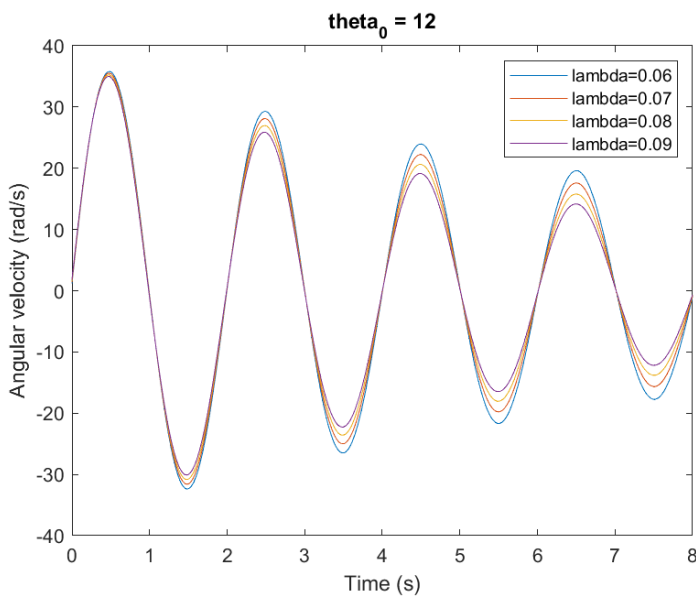


Figure 12: Angular velocity vs time using Equation 11 ($\theta = 4$)

Velocity Analysis

Our objective in this section of the study is to explore how the pendulum's velocity changes over an 8-second period starting from rest. Typically, the graph of angular velocity will initiate at zero because the pendulum starts from a stationary position. As the motion progresses, we expect the amplitude of the velocity to decrease gradually. This reduction is due to the effects of friction or resistance within the system, which acts to dissipate energy, thereby slowing down the pendulum and eventually bringing it to a halt.

To thoroughly investigate this behavior, we use MATLAB to simulate the pendulum's velocity and generate the corresponding oscillation curves. These simulations are based on numerical solutions from Equation 4 and Equation 6, as well as the analytical approach outlined in Equation 11:

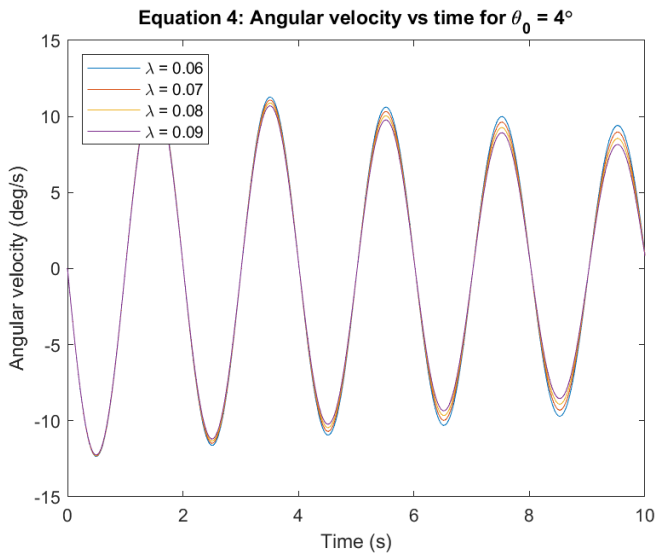


Figure 15: Angular velocity vs time using Equation 4 ($\theta = 4$)

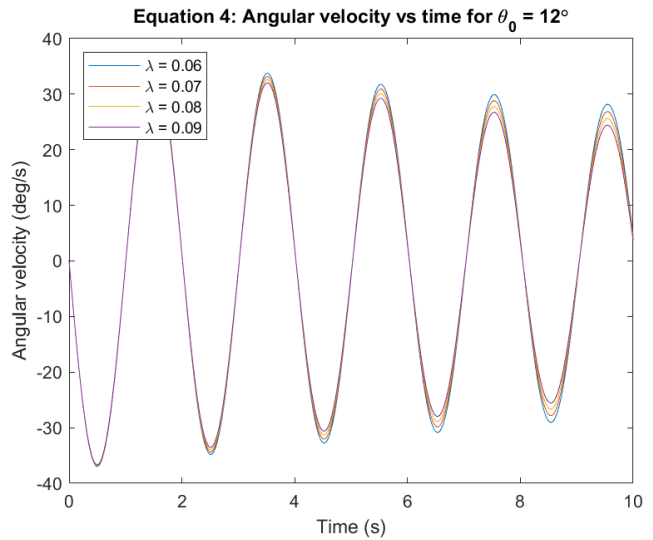


Figure 16: Angular velocity vs time using Equation 4 ($\theta = 12$)

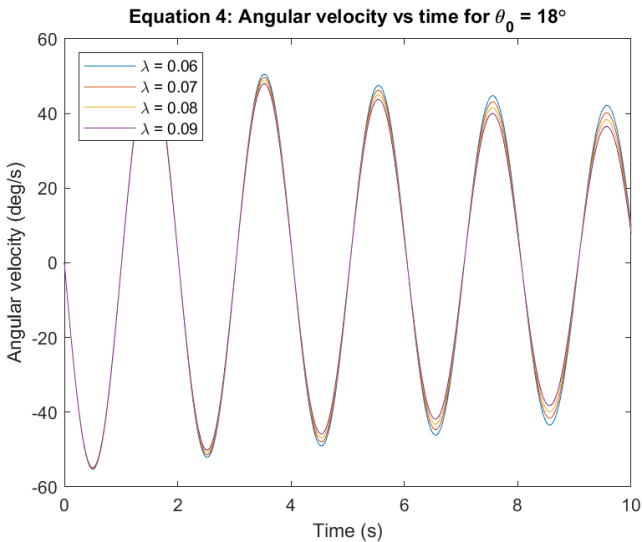


Figure 17: Angular velocity vs time using Equation 4 ($\theta = 18$)

$$\theta(t) = \theta_0 \exp(-\lambda t) \left(\cos(\alpha t) + \frac{\lambda}{2m\alpha} \sin(\alpha t) \right)$$

This equation helps us understand how the initial velocity and the frictional forces impact the pendulum's motion over time. By analyzing the resulting graphs, we can observe the following:

- **Initial Conditions:** The velocity plots all start from zero velocity, adhering to the conditions of being released from a state of rest.
- **Decay of Motion:** Over time, the amplitude of the angular velocity decreases, a trend that graphically represents the loss of kinetic energy due to friction and air resistance encountered by the pendulum.
- **Stopping of Motion:** Eventually, the amplitude of the velocity reduces sufficiently that the pendulum ceases all motion, a state reached when all initial kinetic energy has been converted into other forms of energy (mostly heat) or counteracted by friction.

These insights are crucial for understanding the dynamic behavior of the pendulum in practical settings and provide a basis for further experimental design and hypothesis testing regarding motion under frictional influences

Detailed Analysis of Angular Velocity

In our comparative analysis of the angular velocity graphs derived from Equations 4 and 6, several interesting patterns have emerged. These graphs were generated under varying initial displacements θ_0 (specifically 4° , 12° , and 18°) and different friction coefficients (λ).

Observations from the Graphs:

- **Consistency Across Different Conditions:** Despite changes in the initial displacement, the shape of the angular velocity versus time graph remains relatively unchanged. This consistency suggests that the fundamental characteristics of the pendulum's motion are not heavily influenced by the starting amplitude.
- **Impact of Friction Coefficient:** As expected, a lower friction coefficient results in a higher amplitude of angular velocity. This effect is consistent across all tested initial angles, indicating smoother motion under conditions of reduced friction.
- **Relationship Between Displacement and Velocity:** Interestingly, whenever the angular velocity reaches zero, the angular displacement is at either its maximum or minimum. Conversely, peak angular velocities correspond to moments when the displacement crosses through its equilibrium position. This inverse relationship highlights the pendulum's energy transfer between kinetic and potential forms.

Further Insights:

- **Variable Amplitude:** Although the general shape of the velocity graph does not vary with different initial displacements, the amplitude of the graph does. Higher initial amplitudes generally lead to greater angular velocity amplitudes. This trend underscores the greater energy involved in movements initiated from higher displacements.

- **Gradual Divergence in Amplitude:** While the differences in amplitude due to varying friction coefficients might not be apparent initially, these differences become more pronounced over time as the system evolves and the damping effects of friction accumulate.

Mathematical Formulations Used:

- For Equation 4: $\frac{d^2\theta}{dt^2} + \frac{\lambda}{m} \frac{d\theta}{dt} + g \sin(\theta) = 0$
- For Equation 6: $m \frac{d^2\theta}{dt^2} + k\theta = 0$, where $k = mg$

These equations helped plot the detailed trajectories of angular velocity against time, showcasing how the pendulum's motion is influenced by both mechanical settings and environmental resistances.

Conclusion from Graphical Analysis:

The collected data and plotted graphs confirm that the angular velocity of the pendulum behaves predictably under varying theoretical conditions, with friction playing a significant role in modulating the motion's amplitude. The consistent graph shapes across different initial conditions provide a reliable foundation for further experimental investigations and help in understanding the dynamic stability of pendular movements under different operational scenarios.

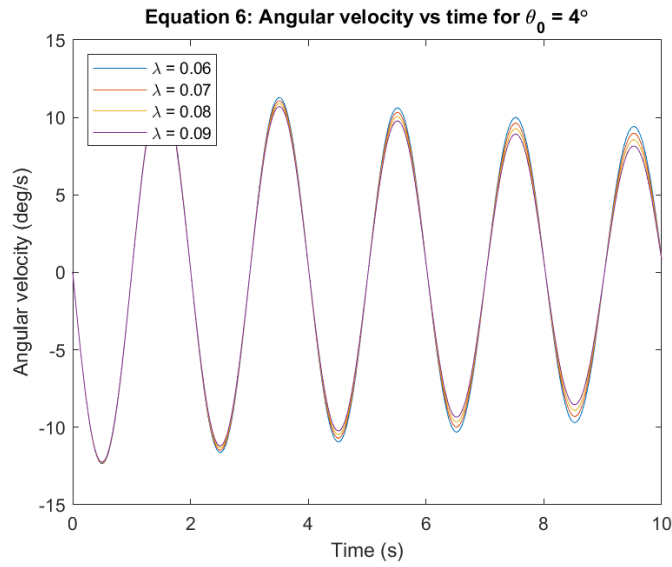


Figure 18: Angular velocity vs time using Equation 6 ($\theta = 4$)

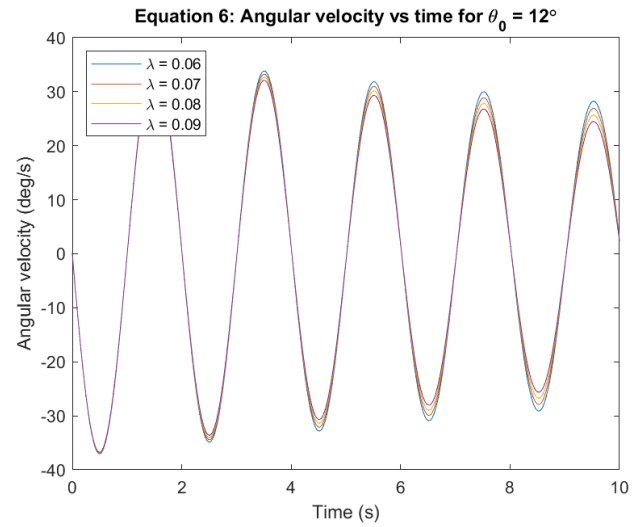


Figure 19: Angular velocity vs time using Equation 6 ($\theta = 12$)

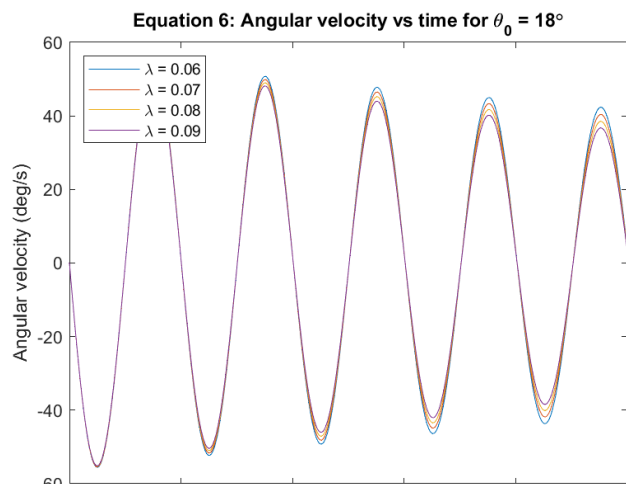


Figure 20: Angular velocity vs time using Equation 6 ($\theta = 18$)

Conclusion

This detailed investigation has provided profound insights into the behavior of a pendulum as influenced by friction and initial setup conditions. It has been distinctly observed that reducing the friction coefficient enhances the amplitude of oscillation, highlighting the critical importance of minimizing resistance to achieve optimal pendulum dynamics.

Furthermore, despite the variations in the initial conditions, the fundamental motion patterns, as depicted by the shape of the displacement and velocity graphs, remained consistent. However, variations in amplitude were clearly dependent on both the initial angle and the level of friction.

Ultimately, this project has not only reaffirmed the foundational principles of mechanical physics but also emphasized the significant practical implications of external factors such as friction on the operational efficiency of pendular systems.

Appendix

Matlab Code Zip files provided separately in the submission